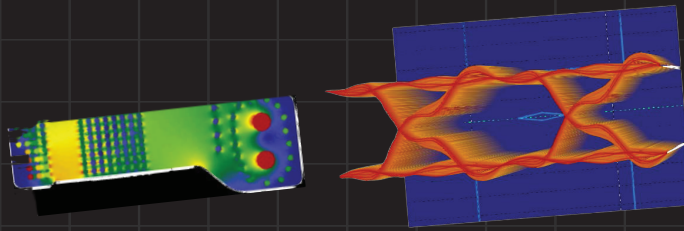
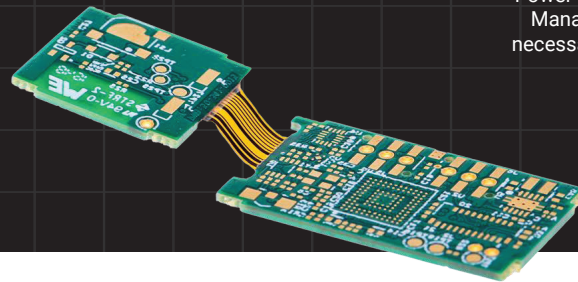


System Engineering



WHY WUERTH ?

- Leading Manufacturer for PCB's in Europe
- Leaders in High end technology PCB Manufacturing
- Leaders in component Manufacturing
- EMI/EMC Specialists
- Quality assurance in accordance to ISO 9001-2015



Wuerth Elektronik India has expertise in the field of PCB Engineering, Manufacturing, and Interconnect solutions. With its vast experience in the field of PCB Domain, engineers engage themselves to Analyze, debug and provide solutions, which will enable the product development teams achieve their goals. In this high-speed digital age, Signal Integrity, Power Integrity and Thermal Management is very much necessary to ensure designs are first time right.

POWER INTEGRITY SERVICES

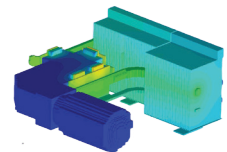
- 3D Voltage distribution plot
- 3D Current distribution plot
- 3D power loss density
- Visualize and list the temperature of planes, pins and vias
- S-Parameter model extraction of PDN
- 3D Field plots and current density
- Self resonant frequencies and quality factor for the PDN 3D field plots.

SIGNAL INTEGRITY SERVICES

- Analysis of connector effects
- Analysis of Eye diagram analysis & optimization of the channel using TX & RX parameters connector effects
- Analysis of Via effects on channel using 3D
- Channel simulator SerDes Modelling using both statistical & bit by bit simulation
- Extracting S-parameters of multi-layer PCB's and Packages

SYSTEM THERMAL MANAGEMENT

- Heat Sink Optimization
- Thermal Performance of an Electronic product
- Enclosure Optimization
- Steady State and Transient Analysis
- Natural and Force Convection



OUR DESIGN EXPERTISE

- SI, PI, and EMC analysis for PCB-level
- Layer stack-up, hybrid technologies, via, and net topology design
- Capabilities/Experience to 40GHz+ and 25Gbps+
- IBIS model simulations, including Power Aware IBIS v5.0
- Expertise in high-speed memory interfaces (DDR3 and DDR4)
- High Power PCB designs upto 15KW.
- Shield designs for both compliance and sensitive signal preservation
- Design review for EMI and recommendations for EMI Filters
- Pre and post-layout, high-speed signal integrity analysis to verify signal quality

- Reducing signal reflections and crosstalk
- Increasing timing margins.
- Enhancing power integrity through cost effective decoupling design, simultaneous switching noise (SSN) reduction and reducing conducted and radiated emissions.
- IC Package level thermal analysis
- Single and Multiple PCB level Thermal analysis
- Complete electronic system level thermal Management for IP XX designs.

TOOLS FOR HIGH END SYSTEM DEVELOPMENT

cadence

ALLEGRO LAYOUT

Altium

ALTUM LAYOUT

SIEMENS

PADS LAYOUT
HYPERLYNX
SI/PI/THERMAL

Ansys

ANSYS ICEPACK

SOLIDWORKS

SOLID WORKS

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